BUILD PACKET (format E)

CSBS-BP-001: AI-Assisted PM Mobile Application

Voice-enabled. Platform-agnostic. Read-only. PM-first.

Packet ID	CSBS-BP-001
Date	2026-09-01
Issued by	Ray Delgado, CSBS instruction hub
Owner	Stan (Winston in corpus files; same person)
Recipient	Claude Code, Codex, or any coding agent
Lane / Tier	04 Factory/Labs. T1 generic licensable IP. CSBS is the T2 drilldown. Churchill Pro Solutions pilot building is T3.
Controlling source	00 Churchill Smart Building Systems - Project Design.md (wins on conflict)
Status	Ready to paste. Phase 0 design first. Phase 1 build only after owner approval.

How to use this packet (for Stan)

Upload this PDF to the coding agent with the nine files listed in Section 5. Tell it: **"Execute BUILD PACKET CSBS-BP-001. Phase 0 first. Stop at every stop condition. Do not start Phase 1 until I approve Phase 0."** One packet, one recipient. Save the agent's output verbatim with model, date, and tools enabled before anyone summarizes it.

Control boundary (non-negotiable, applies to every section): Existing controllers keep PID loops, safeties, interlocks, freeze protection, smoke/fire control, emergency shutdowns, and fallback sequences. Blocked without explicit owner, controls-engineer, and OT-security authority: live BMS connection, credentials, confidential building data, OT-network probing, external customer or vendor outreach, paid tools, network changes, controller or setpoint writes.

1. Objective

Design, then build, a voice-enabled, AI-assisted preventive-maintenance mobile application for engineers and operators who maintain equipment behind existing building management systems (BMS/BAS).

The app is platform-agnostic. Any engineer or company must be able to onboard through a guided sync and setup process that adapts the app to their building, equipment, BMS platform, trades, naming conventions, and PM program with zero code changes.

Version one is PM-first and read-only. It ingests exported BMS data and documents, organizes them, generates prioritized PM tasks, produces evidence-backed fault cards from engineer-approved deterministic rules, drafts CMMS-ready work orders and commissioning packets, and gives the operator a voice and text copilot that answers only from cited evidence.

Version one never writes to a BMS. No setpoint, schedule, sequence, override, or command changes. Existing controllers keep all real-time control.

Working name: {APP_NAME}. Do not pick a product name. That is an owner decision.

2. Who it is for

- **Primary users, daily, in the field:** chief engineers, building engineers, HVAC PM technicians, controls technicians.
- **Secondary users, review and approve:** controls engineers and integrators, commissioning providers, facility directors.
- **Readers, briefs and reports:** property managers, building owners, HVAC service company managers.

Design for the primary user first: someone standing in a mechanical room, gloves on, one hand free, bad lighting, weak or no signal, who needs the next task, the evidence behind it, and a way to capture what they found without typing.

3. What the app must do

The product promise, from the controlling design. For every finding, the app tells the operator: **what appears wrong, why it matters, what evidence supports it, which equipment and spaces are affected, what to check in the field, which trade owns it, and what verification closes it.**

3.1 Onboarding and sync (the agnostic layer)

This is the feature that makes the app work for any company or building. Build it as a guided wizard, resumable, with progress saved at every step.

1. **Company and role setup.** Company profile, users, roles (engineer, technician, reviewer/approver, reader), trades list, CMMS in use or none, work-order field preferences.

2. **Building profile.** Type (commercial, residential high-rise, institutional, other), size, floors, zones, central plant yes/no, occupancy schedule source.

3. **BMS platform and integration path.** Platform picker: Niagara, Johnson Controls Metasys, Siemens Desigo, Schneider EcoStruxure, Automated Logic, Trane Tracer SC/SC+, Honeywell, Daikin Applied / MicroTech, Gree gateway / central controller, other, none. Integration path picker: offline CSV or export (version-one default), historian export, BACnet object list export, read-only connector (locked in version one; shows "requires approval gate").

4. **Manufacturer evidence passport.** One per connected system. Twelve fields, all required before any mapping can be approved: manufacturer and product family; equipment model and serial/version; unit controller model and firmware; supervisor or gateway model and software version; native protocol and approved read-only path; exact manufacturer protocol/point document and revision; point map provenance; units, scaling, enumerations, quality flags; trend/alarm/history availability; write capability (recorded, disabled); service/licensing/remote-access dependencies; human reviewer and review date. (Source: Cycle 002.)

5. **Import.** Point list (CSV, BACnet object export), equipment schedule, alarm history export, trend export, documents (sequences of operation, O&M manuals, submittals, controls drawings, PM history). Accept camera capture of nameplates and drawings. Every import records file name, hash, date, and who imported it.

6. **AI-proposed mapping with human review.** The app proposes equipment identity, location, point role, unit, command vs status, relationships (AHU feeds VAV, VAV serves zone, sensor measures X), Haystack tags, and Brick classes. Every proposal carries a confidence score and a reviewer state: proposed, approved, rejected, needs source. **Raw vendor point IDs are preserved permanently and shown next to the canonical name.** Nothing is treated as truth until a human approves it.

7. **PM program setup.** Pick asset templates per equipment family. Every interval, threshold, and acceptance value is a field the engineer fills in and cites a source for (manufacturer manual and page, project sequence, engineer judgment). The app ships with empty or clearly labeled placeholder templates. **It**

does not ship with invented numbers.

8. **Customization.** Naming conventions, priority weights, checklists, work-order export format (CSV, JSON, PDF, CMMS template), notification rules, brief cadence, voice command aliases.

Acceptance of this layer: two different synthetic platforms (a Trane-style supervisor export and a Gree-style VRF gateway register map) onboard to a working PM program with zero code changes.

3.2 Asset and point registry

Hierarchy: site -> building -> floor/zone -> system -> equipment -> component -> sensor/actuator/meter -> controller/gateway (Cycle 001, Layer A). Each node carries native IDs, canonical identity, source documents, evidence passport link, and change history (renames, controller replacements, equipment swaps) without breaking history.

3.3 PM engine

Triggers: calendar, runtime, condition, event/fault. Generates a prioritized task list per user, per building, per day. Each task carries asset, component, trigger, procedure steps, required readings, safety notes, responsible trade, completion evidence required (photo, reading, voice note, checklist), and verification test. Tracks completion with technician, timestamp, and evidence. Distinguishes preventive, predictive, corrective, and commissioning work.

3.4 Fault evidence cards (deterministic rules first)

Rules are authored or approved by an engineer inside the app, versioned, and cite their basis (sequence, standard, manufacturer document). Each card must contain: points used, time window, persistence, deadband, rule version, confidence, impact, likely causes, field checks, responsible trade, verification test, adjudication state. **A card cannot be created if any evidence field is empty.** Statistical anomaly models are a later phase, off by default, and never the sole basis for a card.

3.5 Prioritization

Transparent scoring, weights visible and editable by the reviewer role, applied in this order: life safety or equipment protection -> occupant comfort and IAQ -> energy and demand cost -> equipment wear and failure risk -> confidence and evidence quality -> ease and cost of correction -> classification (controls change, field repair, engineering question). The score shows its math on the card.

3.6 Work orders

CMMS-ready draft from any task or card: equipment ID, location, symptom, evidence, probable cause, required trade, safety notes, parts if known, verification test. Export CSV, JSON, PDF. Adjudication after work: confirmed, rejected, duplicate, suppressed, unable to verify. Every adjudication feeds a calibration dataset that stays with the building.

3.7 Commissioning and functional-test packets

Pre-functional checklists, functional performance test scripts, trend requests, pass/fail evidence, sign-off fields. Sign-off is a human field. **The app never marks anything "accepted."**

3.8 Operator copilot (voice and text, read-only, tool-bound)

Questions like "why is the third floor hot," "which zones are overridden," "what changed after the chiller repair," "what is due on AHU-2 this week." The copilot answers only through approved tools: trend query, alarm query, document retrieval, PM query, work-order query. Every answer cites the points, time windows, and documents it used. Observed facts are labeled separately from hypotheses. The copilot cannot invent thresholds, intervals, or sequences. If evidence is missing it says so and names the source that would resolve it.

3.9 Voice

- **Push-to-talk by default. No always-on listening.**
- Capture: dictate notes, readings, work-order fields, checklist responses.
- Commands: open asset, start task, next step, take photo, read back step, flag issue, create work order.
- Read-back: spoken task steps and safety notes.
- Transcript preserved verbatim, editable, stored with the task. Original never overwritten.
- On-device recognition for capture and commands so it works with no signal. Cloud speech only as an optional fallback and only under the data-routing policy in Section 6.
- Noisy-environment tuning (fans, pumps, compressors) is a requirement, not a nice-to-have.

3.10 Operator briefs

Daily and weekly brief per building: what is due, what is overdue, new fault cards by priority, open work orders, verification pending, data-quality warnings (stale points, missing trends, bad units). Exportable as PDF.

3.11 Offline-first

Full task execution and evidence capture with no connectivity. Local encrypted store. Sync on reconnect with conflict handling that never silently drops a field-captured record.

3.12 Multi-building, roles, audit

Multiple buildings and companies with strict data isolation. Role-based access. Append-only audit log of every import, mapping approval, rule change, AI query, AI output, human decision, and export.

3.13 Document library

Manuals, sequences, drawings, submittals, PM history, with revision, upload date, and linkage to equipment and points. Retrieval cites page or section.

3.14 Data-quality monitoring

Stale points, missing trends, impossible units, duplicate assets, orphan points, broken relationships, sensor drift flags. Surfaced in the brief and on the asset.

4. Out of version one

- Any write to a BMS. No setpoint, schedule, sequence, override, or command changes. No write code path in the repository.
- Live read-only connector to a building. Design the interface, stub it, lock it behind the approval gate.
- Anomaly detection or machine-learning fault models as a basis for cards.
- Digital twin or simulation.
- Energy-savings claims or dashboards that imply savings without an approved baseline and measurement method.
- Code, LL97, or NYCECC compliance certification. Frame NYC value as operations, PM, fault evidence, commissioning support, and measurement data only.
- Engineering sign-off or commissioning acceptance by the app.
- Customer or vendor outreach, app store publishing, paid services.

Anything drifting toward these items stops with `unsafe_scope_expansion`.

5. Inputs and exact files

Attach with this packet, from the CSBS knowledge pack CSBS-BMS-ENGINEERING-RESEARCH-20260901-V1:

1. 00 Churchill Smart Building Systems - Project Design.md (controlling source; wins on conflict)
2. 00-csbs-knowledge-pack-digest.md
3. Cycle 001 - Field Map and Offline Pilot Foundation - 2026-08-31.md (knowledge model Layers A-D; offline pilot outputs; go/no-go metrics)
4. Cycle 002 - Trane Daikin Gree NYC BMS Foundation - 2026-09-01.md (manufacturer evidence passport; canonical overlay model)
5. Handoffs/02 BMS Controls Architecture Handoff.md (data contract list)
6. Handoffs/03 PM and AFDD Engineering Handoff.md (PM matrix, fault card, work-order, and functional-test structures)
7. Handoffs/04 Semantic Data and Digital Twin Handoff.md (worked AHU/VAV example; use as the seed dataset)
8. Handoffs/05 OT Cybersecurity Handoff.md (threat model coverage list)
9. Skills/building-management-systems-ai/SKILL.md (doctrine and pitfalls)

State of the corpus you must respect: none of the eight frontier-model handoffs has been run. There is no approved PM matrix, fault library, or semantic model yet. Build the framework and the seed dataset. Do not fill the gaps with invented engineering values.

6. Constraints

6.1 Control boundary

Existing controllers keep PID loops, safeties, interlocks, freeze protection, smoke/fire control, emergency shutdowns, and fallback sequences. Blocked without explicit owner, controls-engineer, and OT-security authority: live BMS connection, credentials, confidential building data, OT-network probing, external customer or vendor outreach, paid tools, network changes, controller or setpoint writes.

6.2 No external actions

No live vendor API calls, no credentials, no paid services, no outreach, no store submission, no deployment.

6.3 Engineering values

No hardcoded thresholds, PM intervals, setpoints, or acceptance values anywhere in engine code. All are data fields entered by a human with a source citation. Placeholder templates must be visibly labeled "placeholder, engineer must set."

6.4 Sample data

Synthetic, or from a public dataset with the license recorded. Seed profile A must match the Handoff 04 worked example: one building, two floors, one AHU, at least three VAVs, zones, SAT, RAT, OAT, fan status and command, static pressure, damper, valve, one meter, one alarm, one fault card, one PM task. Seed profile B is a Gree-style VRF gateway register map (indoor/outdoor hierarchy, central controller, gateway) to prove agnostic onboarding.

6.5 Frontier-model routing

The data confidentiality policy (board card CSBS-B03) is unresolved. Build a provider-agnostic model adapter with three modes: public, redacted, local/on-device. Default to local or redacted. **No building data leaves the device to a hosted model unless the policy flag is explicitly set by an admin role.** Log every call.

6.6 AI labeling

Every AI-generated field, mapping, card, or answer is visibly labeled AI-proposed until a human changes it to approved. The label is part of the data model, not just the UI.

6.7 Technology

Cross-platform mobile (iOS and Android) from one codebase. Offline-first local encrypted store. Sync service. Modular BMS adapter interface with offline CSV, BACnet object export, and historian export implemented, and the read-only connector stubbed. Semantic layer supporting Haystack tags and Brick classes side by side with native IDs. Abstracted speech layer, on-device first. Abstracted LLM adapter. RBAC. Append-only audit log. Choose the stack and record it in an architecture decision record. Prefer open-source and boring over clever.

6.8 Design

Sharp, modern, easy. Field rules: large tap targets, high contrast, dark mode, one-hand use, minimal typing, voice-first capture, every screen answers "what do I do next." Evidence is one tap from any finding. No dashboards that hide the source.

7. Deliverables

Phase 0: Design. Stop for owner review before Phase 1.

1. Product spec: this packet refined into screens and flows.
2. Screen map and onboarding flow with wireframes or component sketches.
3. Canonical data model: site, building, floor/zone, system, equipment, component, point, trend, alarm, fault card, PM task, work order, approval event, evidence passport, audit event. Align to Handoff 02 data contracts and Cycle 001 Layers A-D.
4. BMS adapter interface spec with the offline export adapters and the locked connector stub.
5. Semantic mapping pipeline spec with confidence scoring and review states.
6. Voice command grammar and capture flows.
7. Model adapter spec with the three routing modes.
8. Threat model summary covering the Handoff 05 list as it applies to a mobile app with no live connector.
9. Ten architecture decision records.
10. Test plan mapped to Section 8.

Phase 1: Build. Only after Phase 0 approval.

1. Repository with README, setup, and run instructions.
2. Working app on iOS and Android simulators.
3. Both synthetic building profiles onboarded end to end.
4. Onboarding wizard, evidence passport, import, mapping review, asset registry, PM engine, fault cards, prioritization, work orders with export, commissioning packet, copilot, voice capture and commands, briefs, offline sync, RBAC, audit log, document library, data-quality monitor.
5. Automated tests for every check in Section 8.
6. Verification receipt with real command output, not a self-report.

Phase 2: Design only. Do not build.

Read-only connector activation plan and approval gate. Anomaly model integration plan. Digital-twin hooks. Deliver as documents.

8. Acceptance checks

Every check needs real output in the verification receipt.

1. **Agnostic onboarding.** Both synthetic profiles onboard to a working PM program with zero code changes.
2. **Provenance.** Every canonical point links back to its raw vendor ID, source file, hash, and import date. Trace ten random points.
3. **No invented numbers.** Static scan of engine code finds zero hardcoded thresholds, intervals, or setpoints. Placeholder templates render the placeholder label.
4. **Evidence gate.** Creating a fault card or PM task with any empty evidence field fails.
5. **Voice.** Dictate a work-order note hands-free in airplane mode; transcript stored verbatim; edit it; original preserved.
6. **Offline.** Complete a PM task with photo, reading, and voice note in airplane mode; reconnect; sync with zero loss and a visible conflict log.
7. **No write path.** Static scan finds no BMS write service, no connector credentials, no network egress to BMS endpoints. Connector stub returns `blocked_requires_winston_approval`.
8. **Audit.** Every AI query, AI output, mapping approval, rule change, and export produces an audit event. Delete attempts fail.
9. **Prioritization.** Score reproduces by hand from the visible weights on three cards.
10. **Copilot.** Three sample questions return answers with cited points, time windows, and documents. One question with missing evidence returns a named missing source, not a guess.
11. **AI labeling.** Every AI-proposed record shows the label until approved; the label change is an audit event.
12. **Routing.** With the policy flag unset, no building data reaches a hosted model. Network log proves it.
13. **Exports.** Work order exports to CSV, JSON, and PDF. Brief exports to PDF.
14. **Isolation.** A user in Company A cannot read Company B data by any route tested.
15. **Field usability.** Core task flow (open task, read back steps, capture evidence, close) completes with voice and one hand on a phone-size screen.

9. Stop conditions

Stop, label, and wait. Do not invent past a missing source.

- `blocked_requires_winston_decision`: product name, entity that owns the repository, store accounts, paid services, model provider, first test engineer, any UX choice that changes scope.
- `blocked_missing_bms_source`: any real export format, point map, or platform behavior not documented in the attached files or official vendor documentation.
- `blocked_requires_controls_engineer`: before any PM interval, fault rule, sequence, or acceptance value is treated as approved rather than placeholder.
- `blocked_requires_platform_source`: any vendor-specific adapter claim about Trane, Daikin, Gree, or any listed platform.
- `unsafe_scope_expansion`: any write path, control logic, autonomous action, savings claim, or compliance certification feature.

10. Unresolved owner decisions (do not decide these)

From the CSBS board:

1. CSBS-Q01: first building or use case.

2. CSBS-Q02: likely BMS platform.
3. CSBS-Q03: controls-engineering authority.
4. Entity and IP ownership: Apex Prometheus Labs vs the current Churchill Painting Corp filing. Unconfirmed.
5. Frontier-model handoffs 01 through 08 not yet executed.
6. "Darken" = Daikin is assumed, not confirmed.
7. CSBS-B03: model and data confidentiality policy.

New from this packet:

1. App working name.
2. Mobile stack preference, if any.
3. First test engineer who will run the onboarding on a synthetic profile.

11. Independent-review checklist

For whoever reviews the agent's output before it reaches Stan.

- Same project name everywhere.
- No unlabeled placeholders; no TBD or TODO in shipped screens.
- No write path, credentials, or live connector.
- No hardcoded engineering values.
- Every AI output labeled and cited.
- Both synthetic profiles onboard without code change.
- Verification receipt contains real command output.
- Model, run date, exact prompt, and tools enabled recorded for every agent run.
- Contradictions with the controlling design are listed, not smoothed. Mark each AGREES, CONTRADICTS, or UNSUPPORTED.

12. T1 template flag

This packet is T1 generic licensable IP. Placeholders: {APP_NAME}, {COMPANY_NAME}, {BUILDING_NAME}, {BMS_PLATFORM}, {TRADE}, {CMMS}. CSBS is the T2 drilldown. The Churchill Pro Solutions pilot building is T3. Store the generic version in the Labs inventory once the entity decision is made.